Searching Legislative Design Space for Compromise and Productivity

A Computational Framework for Comparing Congress-Style Institutional Designs

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Modern legislatures are often judged by two goals that pull against each other: they should be productive enough to act, and they should be representative enough to convert competing public interests into legitimate compromise. This paper presents a computational framework for searching a broad legislative design space rather than defending one constitutional reform. The simulator compares Congress-style institutional systems under shared assumptions, including ordinary majority rule, supermajority rule, default enactment unless a blocking coalition forms, challenge vouchers, cross-bloc cosponsorship, adaptive procedural tracks, sunset review, earned proposal credits, affected-group harm guardrails, active-law review, and competing policy alternatives.

The framework separates productivity from compromise, public responsiveness, agenda control, and legitimacy-related risk. Legislators are modeled as individual actors with heterogeneous ideal points and voting sensitivities; bills are proposed around agenda setters; enacted bills update a scalar status quo; and institutional procedures are implemented as composable stages. Across 18 assumption cases and 26 institutional scenarios in the current campaign, open default enactment produces high throughput but also high low-support passage, larger policy shifts, and greater proposer gain. Committee guardrails sharply reduce those risks at large productivity cost. Challenge-voucher sweeps reveal a throughput/safety frontier. Cross-bloc cosponsorship gates improve welfare and reduce proposer advantage by requiring coalition breadth before agenda access, but can become blunt screens when applied uniformly. Adaptive procedural tracks preserve more agenda access while routing risky bills into stronger review lanes. Distributional-harm guardrails reduce minority-harm and improve legitimacy metrics. Active-law review makes some risky enactments reversible after delay. Structured mediation and competing-alternative stages show that compromise can come from changing proposal content before final ratification, not only from blocking or passing fixed bills. Richer lobbying scenarios separate organized-interest pressure into direct, agenda, information, public-campaign, litigation, and defensive anti-reform channels. The contribution is a reproducible mechanism-search framework for identifying institutional designs that can improve compromise and productivity while remaining responsive to public support and generated public benefit.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies → Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, compromise, collective decision-making, voting systems

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1 Introduction

The motivating question is not whether one unusual voting rule should replace Congress. It is whether systematic design search can discover legislative structures that better combine productivity, compromise, and public responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because dissent has been suppressed. A useful legislative system should instead help representatives with competing interests reach compromises connected to the preferences and welfare of the people they represent.

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That goal makes raw legislative productivity a poor target by itself. A system that passes many laws may be efficient, but it may also be volatile, captured, or weakly legitimate. A strict consensus system may be cooperative but unable to adapt. A simple majority system may be responsive to a median coalition but inattentive to concentrated minority harm. The core problem is therefore comparative: which institutional designs produce the best tradeoffs among productivity, compromise, representative responsiveness, minority protection, and resistance to agenda capture?

We build an individual-legislator simulation framework to study that question. The simulator compares institutional rules under shared generated worlds and reports a dashboard of outcomes: enacted bills, floor load, welfare proxy, coalition support, low-support passage, policy movement, proposer gain, gridlock, committee rejection, and challenge use. Each final choice is still a yes-or-no vote, but bills may pass through proposal access, committee information, committee gatekeeping, challenge windows, adaptive routing, sunset review, bicameral approval, or executive veto before that vote.

Default enactment is one test case within that broader search. It is useful because it sharply reverses the burden of collective action: proposed laws pass unless a blocking coalition stops them. But the simulator is not designed to validate default enactment as the preferred reform. It is designed to compare default enactment against ordinary thresholds, coalition-breadth gates, scarce challenge rights, proposer accountability, reversible lawmaking, and future systems that may be more radically different. The central thesis is that legislative design should be evaluated as a mechanism portfolio: agenda rights, proposal volume, screening costs, information institutions, coalition-breadth requirements, blocking rights, reversibility, and proposer accountability jointly determine whether a system produces compromise or merely produces output.

2 Related Work

The project sits between computational political modeling, formal legislative theory, and institutional design. Individual-level simulation models are useful when heterogeneous actors interact under rules that create path-dependent aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Prior work has used such models to explore political representation and legislative performance, including simulations of parliament composition and decision efficiency [Pluchino et al. 2011]. We use the method differently: the object of comparison is not the demographic composition of a legislature, but the procedural architecture through which proposals reach enactment.

Formal legislative theory motivates the model structure. Spatial voting motivates one-dimensional policy positions and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989]. Veto-player theory and pivotal-politics models motivate comparing simple majorities, supermajorities, bicameralism, and executive vetoes [Krehbiel 1998; Tsebelis 2002]. Agenda-control theories motivate treating bills that never reach the floor as substantively important, not as missing observations [Cox and McCubbins 2005]. Empirical legislative-effectiveness work further warns that productivity and influence should not be collapsed into a single measure [Volden and Wiseman 2014].

One tested design family, default enactment, has real institutional analogues, although not usually in broad ordinary lawmaking. WTO dispute-settlement reports are adopted unless there is appeal or consensus not to adopt [World Trade Organization 1994]. EU fiscal-governance procedures use reverse qualified-majority mechanisms in some enforcement contexts [European Parliament and Council of the European Union 2011]. Tacit-acceptance treaty amendment procedures and silence procedures allow decisions to take effect unless objections are raised by a deadline [Churchill and Ulfstein 2000]. In the United States, review-and-disapproval structures such as the Congressional Review Act let agency rules

take effect unless Congress and the President complete a disapproval process [Carey and Davis 2024; United States Congress 1996].

These analogues share a design lesson: institutional rules rarely operate alone. Burden-shifted decision rules tend to be paired with constrained proposers, bounded subject matter, information review, objection windows, or post-enactment correction. The simulator therefore treats each scenario as a bundle of mechanisms, not as a single isolated threshold.

3 Model

Each simulation run creates a legislature, a stream of bills, a scalar status quo, and an institutional process. The model is intentionally austere so that procedural effects are visible.

3.1 Actors and Bills

Legislators have ideal points in $[-1, 1]$, party labels, and heterogeneous sensitivities to party, constituency, lobbying, reputation, and compromise. A legislator's vote is generated by a weighted strategy that evaluates the bill relative to the current status quo and to the legislator's own incentive profile.

Bills have a proposer, a policy position in $[-1, 1]$, public support, public-benefit score, lobby pressure, salience, affected-group support, concentrated harm, and compensation cost. Proposals are generated around proposers so that agenda-setter advantage is measurable. When a bill is enacted, the status quo moves to that bill's policy position. Votes are therefore not abstract preferences over isolated proposals; they are evaluations of a policy change.

3.2 Institutional Stages

The implementation separates voting rules from legislative processes. A voting rule determines how a chamber interprets yes and no votes. A legislative process determines what stages a bill must pass through. Current processes include unicameral majority rule, 60 percent supermajority passage, default passage unless two-thirds block, bicameral majority rule, presidential veto with override, proposal-access screens, proposal-cost access, cross-bloc cosponsorship access, committee gatekeeping, committee information review, challenge vouchers, adaptive routing, sunset review, active-law review, earned proposal credits, budgeted lobbying, distributional-harm review, structured amendment mediation, and competing-alternative selection.

This modular design matters because no legislative rule is meaningful in isolation. The same final threshold can behave differently when proposals are open to all legislators, filtered by a committee, priced by agenda costs, routed by risk, exposed to scarce challenge rights, or revised through amendment. The framework is intended to make those combinations easy to test.

3.3 Challenge Vouchers

In the challenge-voucher process, bills start in a default-enactment lane. Parties receive a limited number of challenge tokens per session. Legislators evaluate whether a bill is sufficiently harmful or risky to justify spending a party token. If a challenge is filed, the bill leaves the default lane and is routed to an active affirmative vote. If no challenge is filed, the bill is enacted by default.

The voucher mechanism tests whether default enactment can preserve fast treatment for low-conflict bills while forcing contested bills into a higher-legitimacy lane. It differs from ordinary veto power because objection capacity is scarce. It differs from committee gatekeeping because opponents must actively spend a limited procedural resource.

3.4 Cross-Bloc Cosponsorship

The cross-bloc cosponsorship process sits before floor consideration. A proposer must attract credible support from legislators outside the proposer's party bloc and at a minimum ideological distance. The rule estimates whether potential outside-bloc cosponsors would publicly attach themselves to the bill based on policy fit, public support, generated public benefit, compromise preference, status-quo shift, and lobby-risk signals.

This mechanism tests whether agenda access can reward coalition breadth without relying on a permanent committee veto. It is intentionally weaker than a final vote: a bill still ultimately passes or fails under the downstream institutional process. The gate only asks whether the proposal has enough cross-bloc credibility to deserve the default lane.

3.5 Adaptive Tracks

Adaptive tracks route each bill to a procedural lane according to a generated risk score. The score combines low public support, policy shift, positive lobby pressure, salience, disagreement between public support and generated public benefit, and low generated benefit. Low-risk bills stay in a fast default-pass lane. Middle-risk bills go either to active simple-majority review or to challenge-voucher review, depending on the scenario. High-risk bills go to informed guardrails.

This mechanism tests whether routine and risky bills should receive different procedures rather than being forced through one uniform legislative track. It also exposes a design choice: the middle lane can prioritize stronger consent, or it can prioritize throughput by using scarce challenge rights.

3.6 Reversibility and Law Review

Sunset trial review treats risky enacted bills as provisional. After the downstream process enacts a bill, the sunset wrapper applies an automatic review based on generated public benefit, public support, lobby risk, and the same risk score used by adaptive tracks. If the review score is high enough, the law is renewed and the status quo remains at the enacted policy. If review fails, the trial expires and the status quo rolls back.

The active-law registry extends this idea over later bills in the run. Risky enacted laws are recorded with a review date; when review is due, the registry estimates realized public value, affected-group support, and risk. Failed laws can be repealed, partially rolling back their status-quo movement. This makes delayed correction, low-support active-law share, and status-quo volatility measurable.

3.7 Earned Proposal Credits

Earned proposal credits replace a static proposal-cost screen with stateful agenda rights. Each proposer starts with a small credit balance and receives a modest session drip. Proposals spend credits according to generated risk, positive lobby pressure, salience, and public value. After the downstream process considers the bill, the proposer earns or loses credits based on public benefit, public support, risk, lobby pressure, and whether the bill passed with weak support.

This mechanism tests whether agenda access can learn from proposer track records. It differs from a fixed proposal cost because bad proposals reduce future access for that proposer, while high-quality proposals replenish agenda capacity. It is still intentionally simple: credits are attached to individual proposers, not parties or coalitions, and the current metric suite does not yet report proposer concentration directly.

3.8 Budgeted Lobbying

Budgeted lobbying makes organized interests explicit actors rather than only bill-level pressure. Lobby groups have issue-preference maps, preferred positions, budgets, capture strategies, information bias, public-campaign skill, public-support mismatch tolerance, and defensive multipliers. The lobbying wrapper allocates spending across direct pressure, agenda access, information distortion, public campaigns, litigation or delay threats, and defensive anti-reform pressure. The mechanism is not intended to validate lobbying as desirable. It is included so capture pressure, defensive spending, public-preference distortion, channel shares, and public benefit per lobby dollar can be measured rather than folded into a single voting weight.

3.9 Structured Amendment Mediation

Structured mediation adds a bounded content-change stage before the final yes-or-no decision. When a bill has weak public support, large policy shift, high capture risk, or strong proposer advantage, the mediator moves the bill toward a weighted target combining the chamber median, the current status quo, and the proposer position. The amended bill then proceeds through the downstream institutional process. This does not model full legislative bargaining, but it adds an essential distinction: a system can improve compromise by revising proposal content, not only by passing, blocking, or reviewing fixed proposals.

3.10 Distributional Harm and Alternatives

Distributional-harm processes model the case where aggregate support may coexist with concentrated loss. Harm-weighted thresholds route high-harm bills to a stronger affirmative threshold. Compensation processes reduce concentrated harm and improve affected-group support at some public-benefit cost. A stricter variant denies agenda access when affected-group support remains too low after compensation.

Competing-alternative processes address agenda-setter power directly. Instead of asking the chamber to accept or reject only the proposer's bill, the process generates same-domain alternatives: a median alternative, a compromise alternative, an opposition substitute, and a low-harm alternative. Selection rules can choose the highest-benefit alternative, highest-support alternative, closest-to-median alternative, or pairwise majority winner before final yes-or-no ratification. This lets the simulator test whether compromise improves when the status quo is not the only alternative to the proposer's preferred bill.

4 Metrics

The simulator reports a metric suite rather than a single score:

- Productivity: share of potential bills enacted.
- Floor consideration: share of potential bills reaching the floor.
- Enacted bills per run and floor bills per run: absolute throughput and institutional load.
- Welfare: average generated public-benefit score of enacted bills.
- Cooperation: productivity weighted by enacted support.
- Compromise: support, moderation, and distance from proposer advantage.
- Low-support passage: enacted bills with less than 50 percent yes support.
- Popular failure: high-public-support bills that fail.
- Policy shift: average absolute movement from the status quo.

- Proposer gain: enacted movement toward the proposer's ideal point.
- Challenge rate: share of potential bills diverted from default enactment into active voting.
- Amendment rate and amendment movement: share of potential bills revised by mediation, and average absolute movement before voting.
- Lobby spend and defensive spend: explicit organized-interest spending per potential bill, the share spent against anti-lobbying reform, channel-specific spend shares, and public benefit per lobby dollar.
- Minority harm and legitimacy: concentrated harm weighted by affected-group opposition, and a proxy combining public support, chamber support, affected-group support, and harm penalties.
- Law-review metrics: active-law welfare, reversal rate, correction delay, and low-support active-law share.
- Alternative metrics: selected-alternative distance from the chamber median, proposer agenda advantage, alternative diversity, and status-quo win rate.

The most important distinction is between productivity and compromise. A high-throughput institution may still enact bills with little active support. Conversely, a low-throughput institution may show high welfare among enacted bills but fail to adapt. The current compromise proxy is only a first step: the long-term target is to model representative compromise more directly by measuring public support, public benefit, coalition breadth, movement away from proposer advantage, and protection against concentrated minority harm.

5 Experimental Design

The current campaign, `simulation-campaign-v14`, runs 150 simulations per case, with 101 legislators, 60 base bills per run, base seed 20260428, 18 assumption cases, and 26 institutional scenarios per case. The assumption cases vary polarization, party loyalty, compromise culture, lobbying pressure, constituency pressure, party count, proposal pressure, flooding conditions, capture pressure, and public appetite for anti-lobbying reform.

The institutional scenarios are:

- unicameral simple majority;
- unicameral 60 percent passage;
- default pass unless two-thirds block;
- default pass plus scarce challenge vouchers;
- default pass plus cross-bloc cosponsorship gates;
- adaptive procedural tracks that route bills by generated risk;
- sunset trial legislation with automatic review for risky enactments;
- earned proposal credits;
- explicit budgeted lobbying, transparency, and anti-capture bundles;
- lobbying-channel variants with democracy vouchers, public advocates, blind origin review, and defensive-spend caps;
- structured amendment mediation for simple-majority and default-pass systems;
- distributional-harm thresholds, compensation, and affected-group consent;
- active-law registry review and repeal;
- competing alternatives and pairwise policy tournaments;
- default pass plus informed guarded committee;

Table 1. Selected scenario averages across 18 assumption cases in `simulation-campaign-v14`.

Scenario	Prod.	Enacted/run	Welfare	Low-sup.	Shift	Prop. gain	Challenge
Default pass unless 2/3 block	0.843	86.911	0.497	0.451	0.665	0.657	0.000
Default pass + channel anti-capture bundle	0.837	86.312	0.514	0.432	0.660	0.657	0.000
Default pass + public-benefit alternatives	0.943	97.150	0.552	0.196	0.008	0.005	0.000
Default pass + median-seeking alternatives	0.943	97.187	0.552	0.195	0.004	0.002	0.000
Default pass + pairwise policy tournament	0.549	56.344	0.641	0.042	0.005	0.004	0.000
Default pass + mediated amendments	0.896	92.273	0.492	0.458	0.499	0.509	0.000
Default pass + challenge vouchers	0.671	77.976	0.510	0.393	0.456	0.540	0.398
Default pass + adaptive tracks	0.443	44.900	0.605	0.266	0.216	0.392	0.000
Default pass + compensation amendments	0.844	86.967	0.490	0.450	0.666	0.656	0.000
Default pass + law registry	0.868	89.687	0.492	0.397	0.834	0.600	0.000
Default pass + cross-bloc cosponsors	0.253	24.705	0.687	0.242	0.081	0.235	0.000
Default pass + informed guarded committee	0.222	22.113	0.651	0.218	0.074	0.249	0.000
Unicameral simple majority	0.265	26.289	0.637	0.000	0.102	0.305	0.000
Unicameral 60 percent passage	0.091	8.652	0.709	0.000	0.020	0.148	0.000

Note: Enacted/run is an absolute count. All other reported columns are normalized rates or scores.

The proposal-cost result reported below comes from `simulation-campaign-v2`, which included proposal costs, bicameralism, and presidential veto scenarios. Challenge-token sweeps over party budgets, member tokens, and q-member escalation come from earlier campaigns. The v14 campaign integrates the main mechanism families through lobbying-channel variants so that the headline comparison remains interpretable.

The campaign is deterministic and reproducible from the repository with `make campaign`. The paper is generated with `make paper`.

6 Code Availability

The simulator source code, LaTeX manuscript source, generated campaign reports, and tracked paper PDF are available in the public open-source repository: https://github.com/Jacoba1100254352/congress-institutional-simulator.

7 Results

Table 1 summarizes selected scenario averages across all v14 assumption cases. The table mixes one absolute load measure with normalized metrics: *Enacted/run* is a count of enacted bills per simulated run, while the other columns are rates or scores on a 0–1 scale. Figures 1 and 2 visualize the same tradeoff space.

7.1 Open Default Enactment

Open default enactment is consistently the broad throughput leader among baseline institutions. Across v14, it averages 0.843 productivity and 86.911 enacted bills per run. By comparison, unicameral simple majority averages 0.265 productivity and 26.289 enacted bills per run, while the 60 percent supermajority rule averages 0.091 productivity and 8.652 enacted bills per run.

The throughput gain comes with risk signals. Open default enactment averages 0.451 low-support passage, 0.665 policy shift, and 0.657 proposer gain. In this model, the rule is not merely reducing gridlock; it is also allowing many proposals to take effect without majority support and moving policy substantially toward proposers.

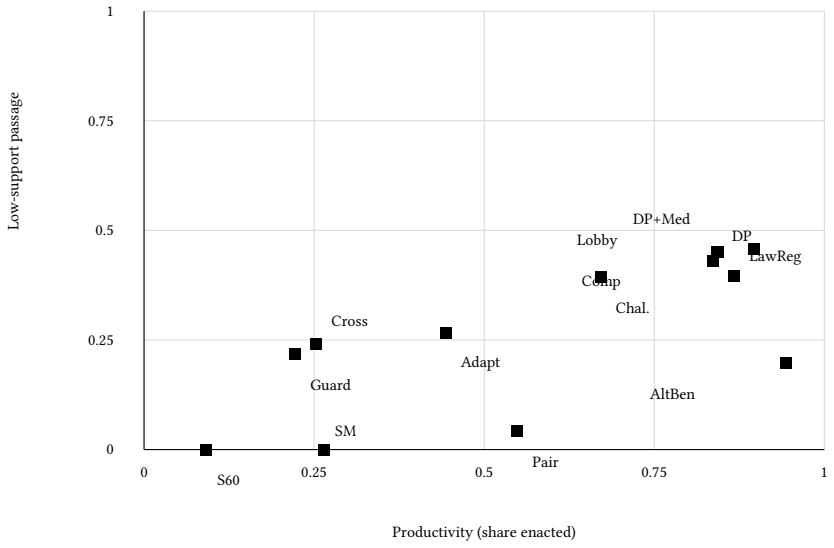

Fig. 1. Productivity versus low-support passage for selected v14 scenarios. The figure makes the main scale distinction visible: productivity and low-support passage are normalized rates, while Table 1 also reports enacted bills per run as an absolute load measure.

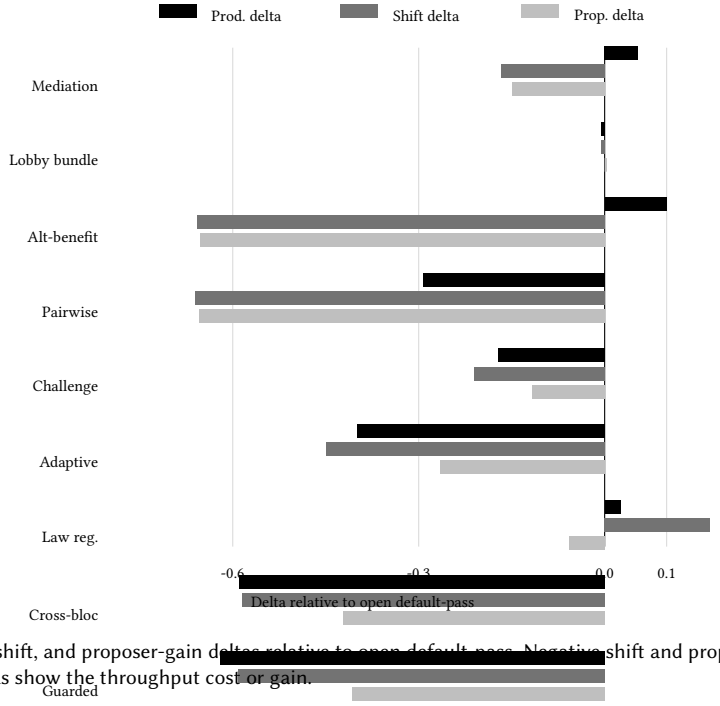

Fig. 2. Productivity, policy-shift, and proposer-gain deltas relative to open default-pass. Negative shift and proposer-gain deltas are desirable; productivity deltas show the throughput cost or gain.

Table 2. Selected challenge-voucher deltas relative to open default enactment.

Case	Prod.	Low-sup.	Shift	Challenge
Baseline	-0.262	-0.065	-0.305	0.500
Low polarization	-0.182	-0.136	-0.142	0.499
High polarization	-0.294	-0.025	-0.361	0.500
Multi-party system	-0.528	-0.290	-0.559	0.815
High proposal pressure	0.018	0.014	-0.037	0.167
Extreme proposal pressure	0.077	0.024	0.022	0.100

7.2 Guardrails

Informed guardrails sharply reduce these risks. Default pass with access screening, committee information, and committee gatekeeping averages 0.222 productivity, 22.113 enacted bills per run, 0.651 welfare, 0.218 low-support passage, 0.074 policy shift, and 0.249 proposer gain. Relative to open default enactment, that is a 0.233 reduction in low-support passage and a 0.591 reduction in policy shift, but also a 0.621 productivity loss.

This result illustrates the central tradeoff. Informational and gatekeeping institutions can make default enactment much safer, but if they are too strong they erase most of the productivity gain that made default enactment attractive.

7.3 Proposal Costs

Proposal-cost screening reduces institutional load but does not automatically improve legitimacy. Across v2, default pass plus proposal costs averages 47.784 enacted bills per run and 50.195 floor bills per run, versus 81.818 and 97.500 under open default enactment. It reduces average policy shift from 0.688 to 0.607.

However, low-support passage rises from 0.455 to 0.491, and proposer gain rises from 0.686 to 1.124. This is not a software bug; it is a substantive mechanism result. If a cost function can be cleared by proposer private gain or favorable lobby pressure, costs may screen out low-value proposals while preserving or amplifying agenda-setter advantage. Proposal costs are therefore an institutional design variable, not a solved safeguard.

7.4 Challenge Vouchers

Challenge vouchers are a promising middle path. Default pass plus challenge vouchers averages 0.671 productivity, 77.976 enacted bills per run, 0.510 welfare, 0.393 low-support passage, 0.456 policy shift, 0.540 proposer gain, and 0.398 challenge use. Relative to open default enactment, it reduces productivity by 0.172, low-support passage by 0.058, policy shift by 0.209, and proposer gain by 0.117.

Table 2 summarizes selected deltas from the earlier challenge sweep. The important failure mode is visible under proposal flooding: fixed token budgets are diluted as proposal volume rises.

The sweep shows that challenge design creates a throughput/safety frontier. Three party-held tokens per party increases average productivity to 0.893 but leaves low-support passage slightly worse than open default enactment. Twenty-five party-held tokens reduce low-support passage by 0.263 and policy shift by 0.478, but productivity falls by 0.454. Member-held tokens and tokenless q-member escalation reduce risk even more, but begin to resemble broad active-vote review. The lesson is not that one voucher design is optimal. The lesson is that challenge rights, token allocation, and escalation thresholds are first-order institutional parameters.

7.5 Cross-Bloc Cosponsorship

Cross-bloc cosponsorship tests agenda access before a bill reaches the default lane. In the default cross-bloc scenario, a bill must attract credible support from at least two legislators outside the proposer's party bloc and at a modest ideological distance. This gate averages 0.253 productivity, 24.754 enacted bills per run, 0.687 welfare, 0.240 low-support passage, 0.080 policy shift, and 0.235 proposer gain.

Relative to open default enactment, the gate reduces floor consideration by 0.744, low-support passage by 0.210, policy shift by 0.583, and proposer gain by 0.421. Earlier cross-bloc sweeps show that stronger gates and cross-bloc-plus-challenge combinations can reduce low-support passage further, but at large productivity cost.

This is an important negative result as much as a positive one. Coalition breadth screens for higher-welfare, lower-volatility bills, but a uniform cosponsorship gate can become too blunt. That motivates adaptive procedural tracks: routine, low-risk bills should not necessarily face the same agenda burden as high-salience, high-uncertainty, or lobby-heavy bills.

7.6 Adaptive Procedural Tracks

Adaptive procedural tracks route bills by a generated risk score rather than applying one admission rule to all proposals. The non-challenge adaptive scenario sends low-risk bills to default enactment, middle-risk bills to active simple-majority review, and high-risk bills to informed guardrails. It averages 0.443 productivity, 44.900 enacted bills per run, 0.605 welfare, 0.266 low-support passage, 0.216 policy shift, and 0.392 proposer gain.

Relative to open default enactment, that reduces productivity by 0.399 but reduces low-support passage by 0.186, policy shift by 0.447, and proposer gain by 0.262. It keeps floor consideration at 0.926, much higher than cross-bloc cosponsorship or informed guardrails. Earlier challenge-middle-lane variants preserve more productivity, but leave higher low-support passage and policy shift. The result is a useful design contrast: challenge rights preserve throughput, while active middle-lane voting removes more weakly supported bills.

7.7 Active-Law Review

Active-law review tests reversibility after enactment rather than agenda exclusion. The law-registry scenario lets default-pass bills enter the floor lane, records risky enactments as provisional laws, and reviews them after delay. It averages 0.867 productivity, 89.565 enacted bills per run, 0.491 welfare, 0.398 low-support passage, 0.835 status-quo volatility, and 0.599 proposer gain.

Relative to open default enactment, the registry increases productivity by 0.026 and lowers low-support passage by 0.052, but it also increases measured status-quo volatility because repeal itself is a policy movement. Its reversal rate is 0.459, average correction delay is 5.127 rounds, active-law welfare at review is 0.585, and low-support active-law share is 0.048. The result supports modeling reversibility as its own tradeoff: correction can reduce the stock of weak laws while increasing churn.

7.8 Earned Proposal Credits

Earned proposal credits are a milder agenda-control mechanism. The base earned-credit scenario averages 0.676 productivity, 64.694 enacted bills per run, 76.341 floor bills per run, 0.513 welfare, 0.441 low-support passage, 0.510 policy shift, and 0.625 proposer gain. It denies 0.202 of potential proposals.

Relative to open default enactment, earned credits reduce productivity by 0.165, floor consideration by 0.202, low-support passage by 0.009, policy shift by 0.153, and proposer gain by 0.031, while increasing welfare by 0.016. This is not a dramatic legitimacy gain, but it is a useful contrast with static proposal costs: the mechanism reduces load without the large proposer-gain increase observed under the earlier cost formula.

Earlier combined earned-credit-plus-challenge runs were more consequential, reducing policy shift and proposer gain much more than the base credit rule alone. The result suggests that proposer accountability works best as one layer in a package, not as a substitute for contestation rights.

7.9 Budgeted Lobbying and Anti-Capture

Budgeted lobbying alone closely tracks open default enactment in v14: productivity is 0.843 and enacted welfare is 0.496, while explicit lobby spend averages 0.103 per potential bill and defensive spend accounts for 0.627 of lobbying spend. This is useful diagnostically even when the headline outcomes barely move. The simulator now distinguishes baseline lobby susceptibility from explicit organized-interest spending, including spending aimed at defeating anti-lobbying reform.

The anti-capture bundle changes outcomes more visibly. Relative to open default enactment, it changes productivity by -0.092, increases welfare by 0.022, reduces lobby capture by 0.090, and reduces private-gain ratio by 0.195. It does not solve capture outright, but it creates a measurable stress test for transparency, public-interest screening, audit sanctions, and lower direct lobby influence.

The new channel-level variants make those diagnostics more specific. A defensive lobbying cap reduces lobby spend per bill from 0.103 to 0.074 and defensive share from 0.627 to 0.381. The equal-access public advocate raises welfare to 0.515 and lowers low-support passage to 0.430. The combined channel bundle has 0.837 productivity, 0.514 welfare, 0.432 low-support passage, 0.229 capture, and 5.065 public-benefit units per lobby dollar. These are modest changes, but they let the simulator test whether anti-capture design should target money in general or particular channels of influence.

7.10 Structured Mediation

Structured mediation adds content revision before voting. In the default-pass mediation scenario, 0.827 of potential bills are amended. Productivity rises from 0.843 to 0.896, policy shift falls from 0.665 to 0.499, and proposer gain falls from 0.657 to 0.509. Compromise increases relative to open default-pass, while welfare changes only slightly.

This result should not be overread as mediation being universally beneficial. Under simple majority, mediation increases productivity but lowers enacted welfare and increases policy movement relative to ordinary simple majority. The more useful finding is conditional: in default-pass settings, a bounded amendment stage can preserve throughput while reducing proposer advantage and policy volatility.

7.11 Distributional Harm and Policy Tournaments

The distributional-harm scenarios show that legitimacy-oriented safeguards can improve minority-protection metrics without always requiring broad front-end gatekeeping. Compensation amendments leave productivity essentially unchanged relative to open default-pass while reducing minority harm by 0.045 and improving legitimacy by 0.019. Affected-group consent is stricter: productivity falls by 0.073, but minority harm falls by 0.046 and legitimacy improves by 0.035.

Competing-alternative scenarios produce the largest reductions in agenda-setter advantage. Public-benefit and median-seeking alternatives both raise productivity to 0.943 while reducing policy shift almost to zero and cutting low-support passage to about 0.196. Pairwise policy tournaments are stricter: productivity falls to 0.550, but welfare rises to 0.641, low-support passage falls to 0.042, and proposer gain falls to 0.004. The obstruction-with-substitute variant behaves similarly, suggesting that forcing a real alternative can be more productive than pure obstruction.

These results are promising but also cautionary. The current alternatives are generated by a stylized procedure with access to chamber medians and public-benefit signals. Future versions should make alternative drafting endogenous, costly, and strategically manipulable before treating policy tournaments as a robust institutional recommendation.

8 Discussion

The results support the broader design premise: a legislative system cannot be judged by passage thresholds alone. Productivity, compromise, public responsiveness, proposer advantage, and minority protection are shaped by the full mechanism bundle. Open default enactment maximizes throughput in the current campaign, but it also increases low-support enactment and proposer advantage. Heavy guardrails reduce those harms but can produce a system less productive than conventional supermajority rule. Proposal costs reduce floor load, but the shape of the cost function determines which proposals survive.

Challenge vouchers illustrate a different design logic. Instead of reviewing every proposal, the institution allocates scarce contestation rights. This preserves a default lane for routine bills while forcing some risky bills into active consent. The v3 through v5 sweeps show that fixed scarce rights can be too scarce under flooding, while broad member-held or tokenless challenge rights can overcorrect into low-productivity review.

Cross-bloc cosponsorship improves proposal access upstream, before a challenge window is needed. Its limitation is that it treats coalition breadth as a universal admission requirement. That is attractive for risky or divisive bills, but wasteful for routine bills with high support and low downside. Adaptive routing responds by making procedure conditional on risk. Active-law review responds from the other side: it lets risky bills pass provisionally, but later reviews and can repeal weak laws. Earned proposal credits add proposer learning, but only modestly change outcomes on their own. Structured mediation and competing alternatives begin to address bargaining over bill content. The next missing layer is endogenous strategy: proposers, opponents, lobby groups, and citizen reviewers should adapt their behavior to the mechanisms rather than passively moving through them.

The broader contribution is methodological. Institutional design proposals often mix multiple mechanisms, making it difficult to know whether outcomes come from thresholds, agenda access, information, bargaining, lobbying, reversibility, or alternative selection. A modular simulator can isolate these components, compare distributions across shared worlds, and identify parameter regions where a design succeeds or fails. The aim is not to crown a winner from the current v14 scenarios; it is to create an expandable search process for discovering and stress-testing more varied legislative frameworks.

9 Limitations

This is not a validated model of the United States Congress or any real legislature. It is a mechanism-exploration simulator. The current model uses one policy dimension, generated legislators, stylized public support, stylized public benefit, simplified lobbying, no elections, bounded mediation rather than full amendment bargaining, generated alternatives rather than strategic drafting, abstract issue domains, and limited bicameral asymmetry.

The largest limitation is endogeneity. Real proposers choose bills strategically based on expected institutional response. The current model mostly generates proposals around proposer ideal points. Default-enactment regimes especially require endogenous proposal generation because the attractiveness of flooding, moderation, and strategic timing depends on the rules.

The second limitation is calibration. Before the simulator can support stronger empirical claims, it should reproduce rough patterns for ordinary legislative systems: coalition sizes, party unity, roll-call ideology distributions, bill progression rates, veto frequencies, and topic-specific throughput.

The third limitation is welfare specification. Public benefit is generated inside the model rather than estimated from external outcomes. It is useful for comparing institutions under common assumptions, but it should not be treated as a real social-welfare measure.

10 Research Roadmap

The next simulations should extend agenda mechanics before adding media, elections, or other realism layers:

(1) add deliberative citizen review or mini-public certification as an independent legitimacy and information screen;
(2) add agenda-scarcity variants such as randomized lotteries, quadratic attention budgets, and public objection windows;
(3) add richer constituent, district, and public-will models so affected-group support is grounded in represented people rather than generated bill fields;
(4) make alternative drafting endogenous, costly, and vulnerable to strategic decoys or cloned alternatives;
(5) deepen mediation into multi-round amendment bargaining with compensation offers for concentrated harms.

These additions directly address the observed failure modes: proposal flooding, low-support enactment, proposer gain, weak compromise formation, and permanent policy movement after weakly supported bills.

11 Conclusion

The current simulations are an early pass through a much larger design space. Default enactment is useful as a stress test because it changes the burden of collective action, but it is only one candidate. Guardrails can reduce low-support enactment, but at large productivity cost. Proposal costs reduce institutional load, but their design determines which bills survive. Challenge vouchers preserve more throughput while reducing some risk metrics, but they become diluted under flooding unless token budgets and allocation rules are tuned. Cross-bloc cosponsorship improves enacted-bill quality and reduces proposer advantage, but can become a blunt gate. Adaptive routing is a more flexible middle path. Active-law review shows that reversibility and correction are separate from front-end screening. Earned proposal credits add stateful proposer accountability and reduce policy shift modestly. Structured mediation, distributional compensation, policy tournaments, and lobbying-channel safeguards show that content revision, alternative generation, and anti-capture channel design can reduce proposer or capture advantage more directly than yes/no screening alone, but the model still needs richer constituent structure, endogenous strategy, and multi-round bargaining.

The paper's core claim is therefore exploratory: computational institutional search can help identify legislative frameworks that better balance compromise, productivity, and representative responsiveness than any single familiar rule or isolated reform proposal.

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